

DL4DS Fine-Grained Visual Classification of Bird Species

Shiheng xu, Renjie Fan, Kunshu Yang

March 1, 2025

Abstract

This project explores Fine-Grained Visual Classification (FGVC) using deep learning, focusing on the CUB-200-2011 dataset. We aim to develop a model leveraging ResNet50 as a baseline, with improvements from attention mechanisms and Supervised Contrastive Learning to enhance classification accuracy and interpretability.

Introduction

Introduction Fine-Grained Visual Classification (FGVC) is an essential task in computer vision with applications in wildlife monitoring, medical diagnosis, and automated quality control. Unlike standard image classification tasks, FGVC requires models to focus on subtle differences between classes, making it more challenging. In this project, we aim to implement a deep learning-based FGVC model using a transfer learning approach with ResNet50 as a baseline. We will then explore enhancements through attention-based mechanisms and Vision Transformers to improve classification accuracy and interpretability.

Related Work

FGVC has been extensively studied in the computer vision community, with several approaches developed to tackle the problem:

CNN-based approaches: Earlier work on FGVC utilized convolutional neural networks (CNNs) with transfer learning, such as ResNet and Swin transformer, achieving moderate success. A convolutional layer processes an input image X to extract hierarchical feature representations:

$$Y(i, j) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} X(i+m, j+n) \cdot K(m, n) \quad (1)$$

where K represents the convolutional kernel and $M \times N$ is its size [1].

Attention mechanisms: Methods like Bilinear CNNs and attention-based networks improve FGVC by focusing on key discriminative regions in the image. Bilinear pooling captures pairwise interactions between feature channels:

$$y = \phi(X) \cdot \phi(X)^T \quad (2)$$

where $\phi(X)$ is the feature representation of the image [2]. Attention mechanisms, such as the Convolutional Block Attention Module (CBAM), apply both spatial and channel-wise attention to refine feature maps [3].

Swin Transformers: Recent work has introduced *hierarchical* transformer architectures for visual tasks, most notably the Swin Transformer [5]. Unlike global self-attention approaches such as ViT, Swin Transformer limits self-attention to local, non-overlapping windows. By shifting the window partitioning across successive layers, it enables information exchange between neighboring windows, progressively capturing both local and global dependencies. Formally, the self-attention mechanism remains:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (3)$$

where Q, K, V are query, key, and value matrices, and d_k is the dimensionality of the key vectors [5]. However, in Swin Transformer, attention is restricted within windowed regions of the image, then strategically shifted in later stages to merge local features hierarchically.

Because Swin Transformers can efficiently model large images by leveraging this shifted window strategy, they have become a competitive alternative to purely convolutional models in many vision tasks. In our work, we will build upon these ideas by exploring hybrid architectures that combine CNN-style local feature extraction with Swin Transformers' ability to refine representations in a hierarchical, self-attention-driven manner.

Proposed Work

Fine-Grained Visual Classification (FGVC) is a challenging task due to minimal inter-class variations and high intra-class diversity. Our proposed approach follows a two-stage framework: first, we establish a convolutional neural network (CNN) baseline using ResNet50, and then we enhance classification performance using a Transformer-based model, Swin Transformer, combined with supervised contrastive learning and localized attention mechanisms. This approach allows for a systematic comparison between CNN and Transformer architectures in FGVC tasks, while also incorporating advanced optimization strategies to improve fine-grained feature discrimination.

In the first stage, we train **ResNet50** as a baseline model to assess the limitations of CNN-based methods in FGVC tasks. CNNs, despite their strong feature extraction capabilities, struggle with global context modeling and often require additional mechanisms for part-based attention. To address these challenges, our second stage replaces ResNet50 with **Swin Transformer**, a hierarchical vision transformer that incorporates:

- **Windowed Self-Attention**, which efficiently captures local patterns while maintaining computational efficiency.
- **Hierarchical Feature Learning**, enabling multi-scale feature representation for better fine-grained classification.
- **Enhanced Global Context Awareness**, mitigating CNNs’ limitations in capturing long-range dependencies.

Furthermore, we introduce two additional enhancements:

- **Supervised Contrastive Learning (SupCon Loss)**: This loss function encourages intra-class compactness and inter-class separation, improving feature discrimination.
- **Part-Aware Local Attention**: This mechanism ensures that the model focuses on the most discriminative object regions, refining feature extraction for FGVC tasks.

To evaluate the effectiveness of our approach, we will measure:

- **Top-1 and Top-5 Accuracy** on CUB-200-2011 and Stanford Cars datasets.
- **t-SNE Visualization** to analyze feature space separability.
- **Attention Heatmaps** to verify whether the model correctly attends to key object regions.

By systematically comparing ResNet50 and Swin Transformer, and incorporating contrastive learning with localized attention, this study aims to demonstrate the advantages of hierarchical transformers in FGVC while enhancing interpretability through attention-based visualization.

Datasets

For this project, we have selected the CUB-200-2011 dataset as the benchmark for training and evaluating our fine-grained visual classification (FGVC) models. This dataset is widely used in FGVC research and provides an appropriate level of complexity to assess the performance of both our baseline (ResNet50) and the optimized model (Swin Transformer with contrastive learning and part-aware attention).

CUB-200-2011 Dataset

The **CUB-200-2011** dataset is a large-scale fine-grained classification dataset consisting of:

- **11,788 images of 200 bird species**, with approximately 30 images per species.
- **Detailed annotations** including species labels, bounding boxes, and part annotations (e.g., beak, wings, tail).
- **Challenges** such as high intra-class variability (same species with different postures and lighting) and minimal inter-class differences (visually similar bird species).

This dataset is chosen because:

- It provides a controlled benchmark for evaluating FGVC models.
- The presence of part annotations enables direct validation of our **Part-Aware Local Attention** mechanism.
- It allows a systematic comparison between CNN (ResNet50) and Transformer-based models (Swin Transformer).

Dataset Preparation and Processing

To ensure fair evaluation and optimal model performance, we follow a structured data preprocessing pipeline:

- **Image Resizing:** All images will be resized to 224×224 pixels to match the input size required by ResNet50 and Swin Transformer.
- **Data Augmentation:** Random cropping, horizontal flipping, CutMix, and Mixup will be applied to increase data diversity and improve generalization.
- **Normalization:** Pixel values will be normalized using ImageNet mean and standard deviation.
- **Train-Test Split:** We use the official dataset split, with 5,994 training images and 5,794 testing images, ensuring consistency with previous research.

Feasibility and Justification

The CUB-200-2011 dataset is a well-established benchmark for FGVC research, ensuring the feasibility of our project. Its moderate size allows for efficient model training on available computational resources while still presenting a meaningful challenge for model evaluation. The availability of part annotations further strengthens our ability to analyze

the effectiveness of attention mechanisms, making this dataset an ideal choice for our study.

Evaluation

- **Metrics:**

- *Top-1 Accuracy* and *Top-5 Accuracy*: These metrics measure the model’s overall recognition accuracy in a fine-grained setting and provide insight into how well the model handles easily confused classes.
- *Confusion Matrix*: Helps analyze the most frequent misclassification types, shedding light on whether the model adequately distinguishes among classes with visually similar features.

- **Expected Results:**

- By fine-tuning a pretrained CNN (e.g., ResNet50), we expect to achieve around **80% Top-1 Accuracy**. Adopting more advanced methods (such as Swin Transformer) could further improve performance, potentially reaching 85%–90%.
- We will also include model visualizations to qualitatively demonstrate which regions of the image the model focuses on when dealing with highly similar classes.

- **Baselines:**

- We will first fine-tune a standard CNN (e.g., ResNet50) with minimal modifications to establish an initial accuracy baseline.
- We will then compare our approach against existing public work or methods on the same dataset (e.g., RA-CNN, MA-CNN), thereby quantifying improvements in recognition accuracy and interpretability.

We will use the dataset’s official training/testing split or set aside a validation subset from the training data in order to **prevent overfitting and tune hyperparameters**. Final performance will be reported via quantitative metrics on the test set (e.g., accuracy) as well as qualitative analyses such as heatmap visualizations.

Timeline

- **Weeks 1–4: Project Planning and Preliminary Research**

- Define the project topic and objectives; conduct an initial literature review.
- Complete the project proposal.

- **Weeks 5–6: Data Preparation and Baseline Setup**
 - Acquire/clean the dataset and split it into training, validation, and testing sets.
 - Implement the initial baseline model (e.g., fine-tuning a standard CNN).
- **Weeks 7–8: Preliminary Experiments**
 - Train the baseline model and record initial results (accuracy, loss curves, etc.).
 - Perform preliminary hyperparameter tuning and fix any data or implementation issues.
 - Milestone check: evaluate feasibility and adjust the project plan if necessary.
- **Weeks 9–10: Model Refinement and Advanced Development**
 - Integrate planned improvements (e.g., attention mechanisms, advanced data augmentation, or Transformers).
 - Continue iterative testing and tuning.
- **Weeks 11–12: Evaluation and Visualization**
 - Finalize evaluation metrics (e.g., confusion matrix, Grad-CAM visualization).
 - Document and analyze experimental results; conduct error analysis.
- **Weeks 13–14: Finalization, Report, and Presentation**
 - Refine model performance; optimize code and inference speed if needed.
 - Draft the final report.
 - Prepare the final presentation and demo.
 - Submit the final project report and all relevant deliverables (code, documentation, etc.).

Conclusion

This project addressed Fine-Grained Visual Classification (FGVC) on the CUB-200-2011 dataset using deep learning approaches. We began with a pretrained ResNet50 model as a baseline, leveraging transfer learning to achieve a solid initial accuracy in a fine-grained setting. Building upon this, we introduced attention mechanisms and Vision Transformer architectures to further enhance the model’s ability to capture subtle differences and improve interpretability.

Our experiments showed that incorporating part-based attention noticeably improves the model’s focus on discriminative local features, while the use of a Vision Transformer demonstrated strong potential for learning robust feature representations. Additionally, we employed visualization techniques such as Grad-CAM to highlight the model’s attention areas, thus providing better explainability for how specific image regions influence classification decisions.

Overall, this project offers valuable insights into deep learning methods for FGVC: we confirm that specialized attention mechanisms and Transformer-based models can deliver meaningful performance gains over a baseline CNN, particularly in distinguishing highly similar classes. Future work could explore more sophisticated region-detection approaches, attribute-level annotation, or the application of these methods to larger and more diverse datasets, potentially extending to medical imaging or industrial quality control scenarios where fine-grained analysis is critical.

References

- [1] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278-2324, 1998.
- [2] Tsung-Yu Lin, Aruni RoyChowdhury, and Subhransu Maji. "Bilinear CNN models for fine-grained visual recognition." *Proceedings of the IEEE International Conference on Computer Vision*, 2015.
- [3] Sanghyun Woo, Jongchan Park, Joon-Young Lee, and In So Kweon. "CBAM: Convolutional Block Attention Module." *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018.
- [4] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. "Attention is all you need." *Advances in Neural Information Processing Systems*, 2017.
- [5] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. "An image is worth 16x16 words: Transformers for image recognition at scale." *arXiv preprint arXiv:2010.11929*, 2020.